

Project Overview: Accelerated Fleet Dispatch & Route Intelligence (AFDRI)

1. Problem Statement & Objective

Logistics dispatchers struggle with slow routing calculations and inefficient vehicle assignment at scale. The objective is to build a high-performance data intelligence application that optimizes routes in real-time using GPU-accelerated computing.

2. Idea & Differentiation

Unlike standard CPU-bound dispatch tools, AFDRI leverages local NVIDIA RTX GPU acceleration via PyTorch CUDA. This allows for massive, instantaneous parallelization of distance matrix calculations, significantly reducing compute time for complex routing.

3. USP & Features

- **GPU Acceleration:** Uses local hardware (RTX 3060) to outperform standard CPU workflows.
- **Dual-Mode Cloud Integration:** Hybrid "Mock vs Live" mode for immediate development and cloud-scale deployment via GCP (BigQuery/GCS).
- **AI-Driven Dispatch:** Integrated Gemini Enterprise Agent for natural language conversational dispatch support.

4. Architecture & Technologies

- **UI:** Streamlit (Dashboard).
- **Optimization:** PyTorch CUDA (GPU) vs. NumPy/Pandas (CPU).
- **Data/AI:** Google BigQuery, Cloud Storage, Gemini API.

5. Code Execution & Results

How it works: The `acceleration_engine.py` script performs distance matrix calculations. The Streamlit app displays a side-by-side performance comparison (Benchmark Tab).

Expected Output: Visual charts showing "Execution Time vs. Number of Orders" where the GPU curve scales linearly compared to the exponential growth of the CPU curve.

6. Presentation Tips

Highlight the live benchmark tab; it proves the value proposition instantly. Explain the transition from simulated GCP data to live production data as a seamless configuration change.